

Unless stated otherwise, each question is of equal value [4 pts]. Show all of your work.

Find the domain and range of the following function.

1. $g(x, y) = \sqrt{x - 2y}$

Evaluate the following limits.

2. $\lim_{(x,y) \rightarrow (1,0)} (4x - y)$

3. $\lim_{(x,y) \rightarrow (1,-2)} \frac{y^2 + 2xy}{y + 2x}$

Use the Two-Path Test to prove that the following limit do not exist.

4. $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$

Find the indicated partial derivative of the following functions.

5. Using $h(x, y) = y^3 \sin 4x$, find h_{xx} and h_{xy} .

Use the Chain Rule to find the following derivative. When feasible express your answer in terms of the independent variable.

6. [6 pts] dz/dt where $z = x^3y - x$, $x = t^2$, and $y = 2t^{-1}$.

Use the function $f(x, y) = x^2 - 4y^2 - 9$ to complete problems 7–9.

7. Compute the gradient of $f(x, y)$ at $P(1, -2)$.

8. Find vectors that give the direction of steepest ascent and descent for f at $P(1, -2)$.

9. Compute the directional derivative of $f(x, y)$ at $P(1, -2)$ in the direction of $\mathbf{u} = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$ (note that $|\mathbf{u}| = 1$).

Use the Second Derivative Test to determine (if possible) whether each critical point of the given function is a local maximum, local minimum, or saddle point.

10. $f(x, y) = 4 + x^3 + y^3 - 3xy$ with critical points:

a. $(0, 0)$

b. $(1, 1)$

Find all of the critical points (there are two) of the following function.

11. $f(x, y) = \frac{x^3}{3} - \frac{y^3}{3} + 3xy$